Course Title	ARTIFICIAL INTELLIGENCE LAB		
Course Code:	CMC381-L	**Semester:**	Fall 2026
Credit Hours:	1 (Lab), 3 contact hours per week (course: 3+1)	**Prerequisite(s):**	CMC371
Instructor:	Faiza Ali (Junior Lecturer)	**Class:**	BS (Computer Science)
Office:	A8	**E-mail:**	faiza.ali@iqraisb.edu.pk
Lab Days:	As per timetable (one 3-hour session per week)	**Consultation Hours:**	As announced on LMS
Lab Room:	Computer Lab (as per timetable)	**Knowledge Group:**	Intelligent Systems (ACM KA: IS)

Course Description:

Artificial Intelligence has emerged as one of the most significant and promising areas of computing. This lab gives hands-on experience of the foundations of AI and its basic techniques in Python. Students implement:

- Intelligent agents, task environments (PEAS) and problem formulation (8-puzzle)
- Uninformed search (BFS, DFS, UCS, DLS, IDDFS) and informed search (Greedy best-first, A*)
- Local search and optimisation (Hill Climbing, Genetic Algorithm) and game playing (Minimax, Alpha-Beta pruning)
- Fuzzy sets, fuzzy operations and a fuzzy logic-based expert system
- Machine learning (Naive Bayes, K-Means) and artificial neural networks (Perceptron, MLP with backpropagation)

Course Objectives:

✓ To introduce the foundations of AI, including rational agents, problem solving by search, knowledge representation and decision making.
✓ To enable students to implement and compare classical AI algorithms (search, optimisation, game playing, fuzzy logic) in Python.
✓ To give hands-on experience with modern AI tools and libraries (NumPy, scikit-learn, PyTorch/TensorFlow) for machine learning and neural networks.

Course Learning Outcomes (CLOs):

	At the end of the course, the students will be able to:	GA	BT Level*
1	Apply fundamental concepts of Artificial Intelligence, including rational agents, search algorithms, and fuzzy logic, to solve real-world problems.	GA2	C-3
2	Leverage modern AI tools and frameworks to design and implement basic algorithms, including search strategies and machine learning models.	GA5	P-2

* BT = Bloom's Taxonomy, C = Cognitive domain, P = Psychomotor domain, A = Affective domain. CLO weight: CLO1 = 60%, CLO2 = 40%. SDG: 9 (Industry, Innovation and Infrastructure).

Mapping of CLOs to GAs		
GAs / CLOs	**CLO1**	**CLO2**
GA1 (Academic Education)		
GA2 (Knowledge for Solving Computing Problems)	**X**	
GA3 (Problem Analysis)		
GA4 (Design/Development of Solutions)		
GA5 (Modern Tool Usage)		**X**
GA6 (Individual and Team Work)		

GA1 (Academic Education) is attained automatically on successful completion of the course.

Week No.	Topic / Contents	Lab No.	CLO
	Agents, Problem Solving and Search		
Week 1	Introduction to AI, Intelligent Agents and Environments • Scope of AI, real-world AI applications, PEAS, types of agents; reflex vacuum-cleaner agent in Python	1	CLO1
Week 2	Problem Formulation and the 8-Puzzle Problem • States, actions, transition model, goal test; measuring performance by tracking node expansion	2	CLO1
Week 3	Uninformed Search: BFS, DFS and Uniform Cost Search • Python implementations; comparing their performance on a graph-based maze	3	CLO1, CLO2
Week 4	Depth-Limited and Iterative Deepening Search • DLS and IDDFS; visualising search patterns with graphical tools	4	CLO1, CLO2
Week 5	**Informed Search: Greedy Best-First and A* Search** • A* for the 8-puzzle with Manhattan distance; comparing Greedy and A*	5	CLO1, CLO2
Week 6	Local Search and Optimisation • Hill Climbing for 8-queens; Genetic Algorithm to optimise a fitness function	6	CLO1, CLO2
Week 7	Open Ended Lab / Project Assigned	–	CLO1, CLO2
	Mid Semester Examination (Week 8)		
	Game Playing, Fuzzy Logic and Machine Learning		
Week 9	Adversarial Search: Minimax and Alpha-Beta Pruning • Minimax for Tic-Tac-Toe; performance improvement with Alpha-Beta pruning	7	CLO1, CLO2
Week 10	Fuzzy Sets and Fuzzy Set Operations • Crisp vs fuzzy sets; union, intersection and complement	8	CLO1
Week 11	Fuzzy Logic Based Expert System • AC temperature control: fuzzification, rule evaluation, defuzzification	9	CLO1, CLO2
Week 12	Supervised and Unsupervised Learning • Naive Bayes for text classification; K-Means clustering on the Iris dataset	10	CLO2

Week No.	Topic / Contents	Lab No.	CLO
Week 13	Artificial Neural Networks: The Perceptron • Perceptron for binary classification on a linearly separable dataset	11	CLO2
Week 14	Multi-Layer Perceptron and Backpropagation • MLP from scratch and with PyTorch/TensorFlow; digit classification on MNIST	12	CLO2
Week 15	Revision / Open Ended Lab / Project Assessment	–	CLO1, CLO2
Week 16	Open Ended Lab / Project Assessment	–	CLO1, CLO2
Week 17	Final Examination	–	CLO1, CLO2

Lab Policies	
Software:	Python 3.10+ with Anaconda (Jupyter Notebook) or Google Colab. Libraries: NumPy, Matplotlib, NetworkX, scikit-learn, PyTorch or TensorFlow (Lab 12).
Lab Task Policy:	Each lab task is completed and demonstrated in the same session and graded on the lab rubrics (R1–R5). Notebooks/programs are uploaded to the LMS before the deadline. Late submissions will be penalised as per departmental policy.
Quiz Policy:	No separate quizzes are held in the lab; understanding is assessed through lab tasks, viva and examinations.
Open Ended Lab:	Smart Campus Delivery Robot project, assigned in Week 7 and carried out in groups of 2–3 students; assessed through demonstration, viva and report in Weeks 15–16.
Attendance:	As per university policy. Missed labs must be completed before the next session.
Plagiarism:	Collaboration is encouraged but each student must write and explain his/her own code. Copied programs (from peers, the internet or AI tools) receive zero marks for all rubrics of that task and may be referred to the HoD for appropriate action.

Prerequisites:

CMC371. Programming skills in Python (variables, loops, functions, lists and dictionaries) and basic probability and linear algebra are expected.

Course Textbook / Reference Books			
S No	**Book Title**	**Author(s)**	**Edition / Publisher**
1	Artificial Intelligence: A Modern Approach	S. Russell, P. Norvig	4th Ed., 2021, Pearson
2	Fuzzy Logic with Engineering Applications	T. J. Ross	4th Ed., 2016, Wiley
3	Machine Learning	T. M. Mitchell	McGraw-Hill
4	Deep Learning	I. Goodfellow, Y. Bengio, A. Courville	2016, MIT Press

Grading Policy

Assessment				
Assessments / CLOs	**CLO1**	**CLO2**	**Total**	**Week**
Lab Tasks (Labs 1–12): 25%	13%	12%	25%	1–14
Mid Semester Exam (MSE): 15%	15%	–	15%	8
Open Ended Lab / Project: 20%	–	20%	20%	7–16
End Semester Exam (ESE): 40%	32%	8%	40%	17
Total	**60%**	**40%**	**100%**	

Weights follow the assessment scheme of the CMC381-L lab manual; the instructor may adjust them according to the departmental policy announced at the start of the semester.